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A Project Phase-1 Report

On

Explainable AI-Based Fair Price Prediction and Cost Anomaly Detection for Sheet Metal Procurement

Submitted in partial fulfillment of the requirements for
M.Tech Project Phase-1

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1 Introduction

1.1 Background

Procurement teams in manufacturing organizations routinely evaluate supplier prices using enterprise resource planning records, quotation files, historical purchase orders, and manual cost sheets. These sources are valuable, but they mainly explain what was paid in the past. They do not automatically explain whether the price was fair, whether it reflected current material and foreign-exchange conditions, or whether the supplier price should be challenged during negotiation.

Sheet-metal procurement is particularly suitable for a structured cost-intelligence framework because part cost can be related to observable engineering drivers. Material grade, sheet thickness, blank size, weight, bend count, hole count, process route, surface finish, supplier region, commodity index, and annual volume all influence part price. A fair-price system therefore needs to combine procurement records with engineering attributes, rather than relying on purchase history alone.

This thesis develops an explainable AI-driven Cost Digital Twin for sheet-metal procurement. The system combines should-cost modelling, ERP cleaning, market adjustment, supervised fair-price prediction, SHAP-style explanation, anomaly detection, clustering, supplier benchmarking, and dashboard-based decision support. The aim is to support procurement review with transparent evidence instead of a black-box prediction.

1.2 Current Manual Process

- A buyer or cost engineer collects the supplier quotation and the latest ERP price.
- Engineering drawings are inspected manually to identify material, thickness, dimensions, weight, finish, and manufacturing complexity.
- Commodity price and exchange-rate movement are checked separately, often outside the costing sheet.
- A should-cost sheet is prepared using material, conversion, overhead, and margin assumptions.
- The buyer compares the supplier price against historical prices, benchmark quotations, and manual should-cost estimates.
- Price-gap review is performed using judgement, spreadsheet formulas, and limited documented explanation.

The manual process is understandable but slow. It is also difficult to repeat consistently across hundreds of parts because the evidence is spread across ERP exports, drawings, spreadsheets, emails, commodity references, and supplier discussions.

1.3 Problem Statement

Problem statement

ERP systems show historical purchase prices, but they do not determine whether a paid price is fair, inflated, anomalous, or aligned with current engineering and market conditions. A procurement decision-support system is required to predict fair price, explain the cost drivers, flag abnormal prices, and estimate qualified savings opportunities.

1.4 Research Gap

The reviewed literature supports manufacturing cost estimation, digital-twin implementation, drawing-based feature extraction, explainable AI, and procurement anomaly detection. However, these areas are often studied separately. Existing methods do not usually provide an end-to-end procurement pipeline that cleans ERP data, joins engineering attributes, calculates an explainable should-cost, predicts a fair market price, detects anomalies, clusters comparable parts, and translates model drivers into procurement language.

The identified research gap is therefore the absence of a domain-specific, explainable Cost Digital Twin pipeline for sheet-metal procurement that connects technical part attributes with real purchase history and produces review-ready negotiation evidence.

1.5 Objectives

Table 1: Research objectives and expected outputs

No.	Objective	Expected output
O1	Develop an explainable Cost Digital Twin for sheet-metal parts.	Should-cost breakdown using engineering, market, conversion, overhead, and margin drivers.
O2	Prepare ERP procurement data for modelling.	Cleaned, anonymized, normalized purchase-history dataset.
O3	Predict ML fair price.	Model output for each part with confidence and driver explanation.
O4	Detect pricing anomalies.	Unsupervised anomaly flags for unusual price or cost patterns.
O5	Cluster comparable parts.	Part groups for benchmark and sourcing review.
O6	Estimate qualified savings opportunity.	Savings only where ML fair price is lower than ERP/current supplier price.

1.6 Scope

The project scope is limited to sheet-metal components such as brackets, mounting plates, covers, panels, and fabricated assemblies. This scope was selected because sheet-metal cost can be represented using drawing-derived engineering attributes and repeatable conversion-cost assumptions.

- Included: ERP cleaning, supplier anonymization, should-cost modelling, market adjustment, fair-price prediction, SHAP/fallback explanations, anomaly detection, clustering, dashboard output, and result export.
- Partially implemented: real ERP purchase history is cleaned, while drawing-level engineering attributes are currently represented by a prototype engineering master.
- Future implementation: replace prototype engineering attributes with CAD/OCR/drawing-derived attributes and rerun validation on a larger real dataset.

2 Literature Survey

The literature survey was organized into four themes: sheet-metal and manufacturing cost estimation, digital-twin cost-control frameworks, explainable AI for cost prediction, and procurement anomaly/ERP intelligence. This grouping supports the thesis argument that a procurement-focused Cost Digital Twin must combine engineering cost logic, historical purchase data, AI prediction, anomaly detection, and explainability.

Table 2: Literature survey matrix

Theme	Main references	Use in this thesis
Sheet-metal cost estimation	ten Brinke; Geiger et al.; Arikan et al.	Justifies drawing-derived cost drivers and should-cost structure.
Digital twin and cost control	Oettl et al.; Khoshkonesh et al.; Chen et al.	Supports modelling the system as a data-driven cost replica and decision-support twin.
Explainable AI	Yoo and Kang; Moawad et al.; Patidar et al.	Supports SHAP-style explanations and transparent model output.
Procurement anomaly detection	Herreros-Martinez et al.	Supports unsupervised anomaly detection and cluster-based purchase review.

2.1 1. ten Brinke - Costing Support and Cost Control in Manufacturing

This thesis proposes a cost-estimation and cost-control architecture for manufacturing, with application in the sheet-metal domain. It defines modules such as cost models, cost determination, cost reports, risk analysis, data analysis, and data tuning. The work emphasizes that cost information should be integrated into engineering and product-development decisions, not treated only as post-production accounting data.

Relevance to project: This supports the Cost Digital Twin foundation. It justifies using structured engineering cost drivers such as material, geometry, process effort, overhead, and cost feedback loops for should-cost estimation.

Review line: This paper supports the need for a structured cost model that can generate explainable cost information during procurement and engineering decision-making.

Pipeline used:

- 1. Review manufacturing cost-control needs.
- 2. Build a generic cost-estimation architecture around manufacturing information.
- 3. Define cost models and cost structures.
- 4. Use cost-determination and reporting modules.
- 5. Demonstrate the prototype in the sheet-metal domain.

Algorithms/methods used:

- Generative cost estimation.
- Variant-based cost estimation.
- Direct costing.
- Activity-based costing.
- Cost structures and cost-control feedback loops.
- Risk analysis, data analysis, and data tuning modules.

2.2 2. Geiger, Knoblach and Backes - Cost Estimation for Large Scale Production of Sheet Metal Parts Using ANN

This paper develops a quotation-costing method for sheet-metal parts using CAD-derived product features and Artificial Neural Networks. It focuses on reducing the time and cost of preparing quotations, especially where detailed tool-layout planning is expensive. The paper uses features such as material, sheet thickness, contours, bends, holes, batch size, and tool-related parameters to estimate progressive-die and manufacturing costs.

Relevance to project: This is highly aligned with your drawing/OCR-to-cost direction. It supports the idea that engineering drawing features can become model inputs for automated cost prediction.

Review line: This paper justifies using drawing-derived cost drivers and ML models for sheet-metal fair-price or should-cost estimation.

Pipeline used:

- 1. Take customer inquiry and CAD model as input.
- 2. Extract geometric and technical features from the CAD model.
- 3. Build a product model containing sheet-metal cost drivers.
- 4. Convert features into technology-specific base units such as machining time, profile length, bends, holes, and batch size.
- 5. Estimate tool and manufacturing costs.
- 6. Generate quotation cost using cost sheets and enterprise cost data.

Algorithms/methods used:

- Feature-based quotation costing.
- CAD feature recognition.
- Analytical geometric analysis.
- Semantic networks.
- Artificial Neural Networks for progressive-die tool-cost estimation.
- Quickprop/FlexNet ANN training approach.
- Job-order costing and workcenter costing.

2.3 3. Oettl, Eckart and Schilp - Cost Estimation Approach of a Digital Twin Implementation in Industry

This paper proposes a methodology for estimating the cost of implementing a digital twin in an industrial setting. It considers the company's current digitalization level, required IT infrastructure, hardware, software, implementation effort, running costs, indirect costs, and digital twin application area. The approach is implemented as an Excel-based tool and validated through an industry use case.

Relevance to project: This supports the digital-twin framing of your work and shows that digital twin systems can be studied through structured cost factors and implementation logic.

Review line: This paper supports the framing of my system as a cost-oriented digital twin that combines data, models, and decision support.

Pipeline used:

- 1. Identify digital twin application areas.
- 2. Derive digital twin requirements.
- 3. Map requirements to required IT infrastructure.
- 4. Identify cost types for implementation and operation.
- 5. Assess company digitalization level using checklists.
- 6. Calculate implementation cost using parameters and maturity factors.
- 7. Validate the approach using an industry use case.

Algorithms/methods used:

- Total Cost of Ownership approach.
- Requirement analysis.
- Checklist-based digital maturity scoring.

- Parameterized cost calculation.
- Cost-type classification into investment, implementation, running, and indirect costs.
- Excel-based estimation tool.

2.4 4. Yoo and Kang - Explainable Artificial Intelligence for Manufacturing Cost Estimation and Machining Feature Visualization

This paper proposes a manufacturing cost prediction process using 3D CAD models and explainable AI. It uses deep learning to predict CNC machining cost and applies 3D Grad-CAM to visualize which machining features increase cost. The key contribution is not only cost prediction, but also explaining the geometric features responsible for the prediction.

Relevance to project: This supports your XAI requirement. It helps justify why your model should explain cost drivers such as weight, thickness, bends, holes, material, and surface finish.

Review line: This paper supports using explainable AI so that cost predictions are not black-box outputs but are connected to visible manufacturing features.

Pipeline used:

- 1. Collect 3D CAD models, material data, volume data, and cost/price data.
- 2. Convert CAD models into mesh, voxel, or point-cloud representations.
- 3. Normalize cost and volume data and encode material data.
- 4. Train deep-learning models for manufacturing cost prediction.
- 5. Select the best-performing 3D model.
- 6. Use visualization to explain which machining features increase cost.

Algorithms/methods used:

- 3D Convolutional Neural Network.
- PointNet and VoxNet baselines.
- 3D Grad-CAM for explainability.
- Min-max normalization and log normalization.
- One-hot encoding for material.
- Adam optimizer.
- MSE, MAE, and MSLE loss functions.

2.5 5. Moawad et al. - Explainable AI for a No-Teardown Vehicle Component Cost Estimation

This paper presents a top-down, data-driven component cost estimation approach using machine learning and Shapley-value reasoning. Instead of relying only on physical teardown or bottom-up component costing, the paper estimates how vehicle features contribute to total vehicle price. It uses explainability to allocate price contribution across components and features.

Relevance to project: This supports SHAP-style cost-driver explanation. It is relevant because your system also needs to explain why a predicted fair price is higher or lower.

Review line: This paper supports the use of Shapley/SHAP explanations to convert black-box price prediction into understandable cost-driver contribution.

Pipeline used:

- 1. Collect large-scale vehicle attribute and price data.
- 2. Clean, integrate, transform, and engineer vehicle features.
- 3. Cluster vehicles into comparable market groups.
- 4. Train a vehicle price prediction model.
- 5. Validate model performance using residual and prediction-error analysis.
- 6. Use Shapley-value reasoning to allocate price contribution to features/components.

Algorithms/methods used:

- CatBoost gradient-boosted decision trees.
- Hierarchical clustering for vehicle grouping.
- t-SNE for cluster visualization.
- Shapley values / SHAP-style feature attribution.
- Cross-validation and residual analysis.

2.6 6. Patidar, Mishra, Jain et al. - Transparency in AI Decision Making: A Survey of Explainable AI Methods and Applications

This survey explains why AI systems need transparency, trust, accountability, and fairness. It discusses XAI methods such as LIME, SHAP, interpretable machine-learning models, attention mechanisms, layer-wise relevance propagation, and counterfactual explanations. It also highlights the tradeoff between accuracy and interpretability.

Relevance to project: This gives general XAI theoretical support. Since procurement users need to trust model outputs before using them for supplier negotiation, explainability is a necessary part of the system.

Review line: This paper supports the decision to include explainability in the procurement dashboard and not present only a predicted number.

Pipeline used:

1. Define the black-box problem in AI decision-making.
2. Categorize explainable AI methods.
3. Compare XAI methods by interpretability, complexity, and domain applicability.
4. Review real-world applications in healthcare, finance, criminal justice, and human-computer interaction.
5. Identify future challenges such as accuracy-interpretability tradeoff and ethical AI.

Algorithms/methods used:

- LIME.
- SHAP.
- Layer-wise Relevance Propagation.
- Attention mechanisms.
- Counterfactual explanations.
- Interpretable models such as decision trees, linear models, and rule-based systems.
- Feature-importance and surrogate-model explanations.

2.7 7. Arikan et al. - Machine Learning-Based Manufacturing Cost Prediction from 2D Engineering Drawings via Geometric Features

This recent paper proposes an ML framework for estimating manufacturing cost directly from 2D engineering drawings. It extracts around 200 geometric and statistical descriptors from 13,684 DWG drawings across 24 automotive part groups. Gradient-boosted models such as XGBoost, CatBoost, and LightGBM achieve about 10 percent MAPE, and SHAP is used to identify important geometric cost drivers.

Relevance to project: This is one of the strongest recent supports for your next phase. It directly supports drawing-to-cost automation, geometric feature extraction, explainable prediction, and ERP-integrated cost decision support.

Review line: This paper supports my next step of replacing synthetic engineering fields with OCR/CAD-derived drawing features for automated should-cost template generation.

Pipeline used:

- 1. Collect 2D DWG engineering drawings and historical cost/price records.
- 2. Extract geometric and statistical descriptors from drawings.
- 3. Build drawing-level feature vectors.
- 4. Train gradient-boosted machine-learning models.
- 5. Evaluate prediction accuracy across automotive part groups.
- 6. Use SHAP to explain which drawing features drive cost.
- 7. Position the approach as a deployable CAD-to-cost and ERP-integrated decision-support pipeline.

Algorithms/methods used:

- XGBoost.
- CatBoost.
- LightGBM.
- Gradient-boosted decision trees.
- SHAP explainability.
- Geometric and statistical feature extraction from 2D drawings.
- MAPE-based model evaluation.

2.8 8. Herreros-Martinez et al. - Applied Machine Learning to Anomaly Detection in Enterprise Purchase Processes

This paper applies unsupervised machine learning to anomaly detection in real enterprise purchase datasets. It uses exploratory data analysis, z-score, DBSCAN, K-Means, Isolation Forest, and ensemble prioritization to identify suspicious transactions for audit or purchase-specialist review. It also proposes explainability methods such as LIME, Shapley, and SHAP to help business users understand anomaly candidates.

Relevance to project: This supports your ERP pipeline, anomaly detection, K-Means clustering, and Isolation Forest logic. It also supports explaining anomalies to procurement users.

Review line: This paper supports the use of ML to detect unusual purchase patterns and prioritize procurement cases for review.

Pipeline used:

- 1. Collect real enterprise purchase datasets.
- 2. Perform exploratory data analysis.
- 3. Apply univariate and multivariate anomaly-detection methods.
- 4. Generate candidate anomalous transactions.
- 5. Prioritize candidates using an ensemble approach.

- 6. Add explainability methods so purchase specialists can understand why a case was flagged.

Algorithms/methods used:

- z-score outlier detection.
- DBSCAN.
- K-Means.
- Isolation Forest.
- Silhouette index.
- Ensemble prioritization of anomaly candidates.
- LIME, Shapley, and SHAP as explanation methods.

2.9 9. Khoshkonesh, Mohammadagha and Ebrahimi - Integrated 4D/5D Digital-Twin Framework for Cost Estimation and Probabilistic Schedule Control

This recent paper proposes an integrated 4D/5D digital-twin framework for construction cost and schedule control. It combines BIM, NLP, reality capture, computer vision, Bayesian risk modeling, Monte Carlo simulation, and reinforcement learning. Although the application is construction, the paper is relevant because it shows an end-to-end digital twin workflow that maps documents to cost items, updates cost/schedule status, and supports predictive decision-making.

Relevance to project: This supports the broader idea of automated document-to-cost mapping and digital-twin-based cost control. It is useful for defending your pipeline as more than a standalone app.

Review line: This paper supports the idea that digital twins can integrate documents, cost structures, AI models, and decision dashboards into one cost-control workflow.

Pipeline used:

- 1. Use BIM as the digital project model.
- 2. Apply NLP to map contract/specification documents to standardized cost items.
- 3. Use photogrammetry and LiDAR to capture field progress.
- 4. Align captured progress with BIM to calculate earned value.
- 5. Use computer vision to detect activity completion.
- 6. Update probabilistic schedule forecasts.
- 7. Use reinforcement learning for adaptive resource allocation.
- 8. Provide a 4D/5D digital-twin decision sandbox.

Algorithms/methods used:

- Transformer-based NLP.
- BIM-based 4D/5D modeling.
- Photogrammetry and LiDAR reality capture.
- Computer vision for activity recognition.
- Bayesian inference.
- Monte Carlo simulation.
- Probabilistic CPM.
- Deep Reinforcement Learning.
- Earned Value Management.

2.10 10. Chen et al. - Real-Time Decision-Making for Digital Twin in Additive Manufacturing with Model Predictive Control Using Time-Series Deep Neural Networks

This recent Journal of Manufacturing Systems paper presents a digital twin framework for real-time decision-making in additive manufacturing. It uses a time-series deep neural network as a surrogate model inside a Model Predictive Control framework. The system predicts future manufacturing states and adjusts process parameters to improve quality and reduce defects.

Relevance to project: This paper strengthens the modern digital-twin and manufacturing-AI foundation. It does not focus on procurement cost directly, but it supports the idea that a digital twin should combine prediction, monitoring, and decision support.

Review line: This paper supports the future direction of moving from a static cost model toward a more dynamic digital twin that can update predictions and decisions as new manufacturing or market data arrives.

Pipeline used:

- 1. Collect additive-manufacturing process data.
- 2. Train a time-series deep neural network surrogate model offline.
- 3. Use the surrogate model inside a Model Predictive Control framework.
- 4. Predict future process states such as melt-pool temperature and depth.
- 5. Optimize control inputs such as laser power.
- 6. Apply the first control action to the physical/simulated manufacturing process.
- 7. Feed the observed process response back into the digital twin for the next control cycle.

Algorithms/methods used:

- Time-Series Dense Encoder deep neural network.
- Model Predictive Control.
- Constrained optimization.
- Finite Element Analysis simulation for the manufacturing process.
- PID controller as a comparison baseline.
- Directed Energy Deposition additive-manufacturing case study.

Overall Gap Derived From Literature The literature supports cost estimation, digital twin implementation, explainable AI, drawing-based feature extraction, and purchase anomaly detection. However, these are usually treated separately. The identified gap is the lack of an end-to-end procurement-focused pipeline that converts ERP and technical inputs into a traceable should-cost template, predicts a fair price, detects abnormal pricing, and explains the result in procurement language. ## Project Contribution

The proposed work addresses this gap by combining:

- ERP cleaning and purchase-history preparation.
- Traditional should-cost template logic converted into Python.
- Engineering cost drivers for sheet-metal parts.
- ML fair-price prediction anchored by should-cost.
- Isolation Forest and K-Means for anomaly and similarity analysis.
- SHAP or feature-importance explanation for procurement users.
- A dashboard and Excel/template output for review and validation.

2.11 Overall Literature Gap

The literature confirms that cost estimation, digital-twin design, drawing-based ML prediction, explainable AI, and anomaly detection are individually valid. The remaining gap is integration. A procurement user needs one pipeline that converts ERP and technical inputs into a traceable should-cost template, predicts a fair price, detects abnormal pricing, and explains the result in a language suitable for sourcing review.

2.12 Literature Synthesis for Proposed Work

Synthesis area	Literature implication	Design decision in this thesis
Cost model foundation	Manufacturing cost-estimation literature supports decomposition into material, process, labour, overhead, and margin.	The Cost Digital Twin uses explicit cost components instead of predicting price from ERP history alone.
Drawing feature relevance	Sheet-metal and drawing-based ML studies show that geometry and material attributes are important cost drivers.	The model requires material, thickness, length, width, weight, bends, holes, and finish.
Digital twin framing	Digital twin literature supports connecting data, models, and decision interfaces.	The proposed system is framed as a cost-focused digital twin for procurement decisions.
Explainability	XAI literature shows that model transparency improves trust and auditability.	The dashboard reports SHAP/fallback explanations in procurement language.
Anomaly detection	Enterprise purchase-process studies support unsupervised methods for prioritizing suspicious transactions.	Isolation Forest is used as a review-priority signal rather than a final judgement.
Validation	Applied costing and ML literature require both numerical metrics and domain review.	The final plan combines MAE/MAPE/RMSE/R2 with expert should-cost template validation.

The synthesis shows why a hybrid method is appropriate. A purely rule-based should-cost sheet is transparent but may not learn from purchase history. A purely ML-based predictor may learn historical supplier inefficiency and may be difficult to defend in negotiation. The proposed framework therefore uses the Cost Digital Twin as an explainable engineering anchor and the ML model as a fair-price learning layer.

The literature also supports the decision to treat anomaly detection as a prioritization tool. Procurement anomalies are not always errors; they may be caused by volume, delivery urgency, tooling, quality requirements, supplier region, or one-time commercial conditions. For that reason, this thesis does not label all anomalies as savings. Instead, anomaly outputs are routed to procurement review.

3 System Requirements Specification

3.1 Functional Requirements

Table 3: Functional requirements

ID	Requirement	Description
FR1	Upload or load sheet-metal part data.	System must accept the engineering attributes required for should-cost and ML prediction.
FR2	Clean ERP purchase data.	System must remove invalid rows, normalize fields, anonymize suppliers, and produce a cleaned dataset.
FR3	Compute should-cost.	System must calculate material, conversion, overhead, adjustment, and margin components.
FR4	Predict fair price.	System must run supervised regression models and produce ML fair-price output.
FR5	Explain prediction.	System must identify top model driver through SHAP or fallback feature importance.
FR6	Detect anomalies.	System must flag unusual price or cost patterns.
FR7	Cluster similar parts.	System must segment comparable parts for benchmark review.
FR8	Estimate savings.	System must calculate qualified savings only when ML fair price is below ERP price.
FR9	Display dashboard.	System must provide portfolio, part, supplier, ERP, AI, and regional views.

3.2 Non-Functional Requirements

Table 4: Non-functional requirements

ID	Requirement	Description
NFR1	Explainability	Outputs must be understandable to procurement users and reviewers.
NFR2	Repeatability	The same input data should produce consistent costing and ML outputs.
NFR3	Traceability	Each output should be traceable to ERP, engineering, market, and model inputs.
NFR4	Privacy	Supplier names should be anonymized in processed outputs.
NFR5	Usability	Dashboard must support portfolio scan and part-level drill-down.
NFR6	Deployability	The prototype should run locally and be ready for Streamlit Community Cloud deployment.
NFR7	Extensibility	Architecture should allow future CAD/OCR integration and larger ERP datasets.

3.3 Dataset Requirements

Table 5: Dataset fields required for modelling

Dataset	Required fields	Purpose
ERP	Part Number, Description, Supplier, Country, PO Date, Quantity, Unit Price, Currency	Purchase history and spend baseline.
Engineering	Material, grade, thickness, length, width, weight, bends, holes, finish, part category	Cost drivers and ML features.
Market	Steel index, USD/INR FX, source status, source date	Market adjustment and confidence.
Supplier	Supplier region, labour rate, energy rate, machine rate, overhead, margin	Regional costing and benchmark context.
Validation	Expert should-cost fields and reviewer notes	Field-by-field correctness check.

4 Proposed Methodology

4.1 Overall Architecture

ERP data pipeline	Engineering layer	Market intelligence
Cost Digital Twin	AI pricing layer	Decision dashboard
Clean and anonymize purchase records	Material, geometry, weight, bends, holes, finish	Steel index and USD/INR context
Should-cost and template adjustments	ML fair price, SHAP, anomalies, clusters	Savings, benchmarks, and part drill-down

Figure 1: Overall proposed Cost Digital Twin architecture

The proposed architecture separates procurement data preparation, engineering-cost modelling, market adjustment, machine-learning prediction, anomaly detection, and user-facing analytics. This separation is important because each layer has a different source of uncertainty. ERP data can contain purchase noise, engineering data can contain missing drawing attributes, market data can be unavailable, and ML predictions must be explained before they can be used in procurement negotiation.

Table 6: System module mapping

Module	Role	Prototype artifact
ERP Data Pipeline	Clean and anonymize purchase records	ntp/erp_pipeline.py
Engineering Layer	Provide drawing-derived cost drivers	data/sample_parts.csv; future CAD/OCR attributes
Market Intelligence	Steel index and FX adjustment	ntp/market_data.py
Cost Digital Twin	Should-cost calculation	ntp/cost_model.py
AI Pricing Layer	Fair price, confidence, explanation	ntp/ml_models.py
Decision Dashboard	Portfolio and part-level review	app.py

4.2 Data Flow

Raw ERP	Clean ERP	Cost Twin	ML Label	AI Models	Dashboard
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Figure 2: ERP and engineering data flow

- Raw ERP purchase rows enter the cleaning pipeline.
- The pipeline removes invalid rows, normalizes currency-related fields, maps regions, and anonymizes suppliers.
- Engineering attributes are joined at part level.
- Market data provides steel-index and FX adjustment inputs.
- The Cost Digital Twin calculates an explainable should-cost.
- The ML layer prepares fair labels and trains fair-price prediction models.
- Anomaly and segmentation models produce review-priority signals.
- The dashboard presents portfolio metrics, part detail, model explanation, supplier benchmark, and regional costing.

4.3 Cost Digital Twin Model

Material	Energy	Labour	Bends/Holes	Finish	Overhead	Adjustments	Margin
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Figure 3: Cost Digital Twin cost-component structure

Formula

Should Cost = Material Cost + Energy Cost + Labour Cost + Bend/Hole Complexity Cost + Surface Finish Cost + Overhead + Manual Template Adjustment Cost + Supplier Margin

Formula

Market Adjusted Steel Rate per kg = Base Steel Rate per kg x (Latest Steel Index / Base Steel Index) x (Latest USD/INR / Base USD/INR)

Formula

Qualified ML Savings = max(ERP Price - ML Predicted Fair Price, 0) x Annual Volume

4.4 ML Fair-Price Label Strategy

Raw ERP Price	Gap vs Should-Cost	Clip / Lift Label	Sample Weight	Fair-Price Label
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Figure 4: ML fair-price label preparation logic

Raw ERP price is treated as noisy evidence rather than direct truth. A historical paid price may include urgent buying, one-off tooling, freight, low-volume effects, negotiation weakness, returns, credits, or supplier inefficiency. The fair-price label pipeline compares ERP price with the should-cost anchor, clips abnormal high labels, lifts unusually low labels, and downweights outlier labels before model training.

4.5 Algorithm Stack

Linear Regression	Random Forest	XGBoost/Fallback	SHAP	Isolation Forest	K-Means
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Figure 5: Algorithm stack for prediction, explanation, anomaly detection, and clustering

Table 7: Algorithm implementation status

Algorithm	Role	Status
Linear Regression	Baseline fair-price model	Implemented
Random Forest	Non-linear fair-price benchmark	Implemented
XGBoost / Gradient Boosting fallback	Primary tabular model	Implemented
SHAP / feature importance fallback	Prediction explanation	Implemented
Isolation Forest	Unsupervised anomaly detection	Implemented
K-Means	Part segmentation	Implemented

4.6 Methodological Design Choices

Design choice	Reason	Tradeoff
Sheet-metal-only scope	Improves modelling consistency because comparable drawing attributes are available.	Limits generalization to other manufacturing categories.
Should-cost anchor	Provides explainable engineering reference before ML prediction.	Requires accurate assumptions for material, labour, energy, overhead, and margin.
Cleaned fair-price labels	Avoids training directly on noisy ERP prices.	Label preparation must be transparent and defensible.
Multiple model stack	Allows baseline, non-linear benchmark, and primary model comparison.	Small datasets can overstate training performance.
SHAP/fallback explanation	Keeps explanations available even when SHAP runtime is unavailable.	Fallback feature importance is less local than SHAP.
Unsupervised anomaly detection	Works without labelled procurement anomalies.	Requires expert review before business action.
Dashboard output	Supports portfolio scan and part-level evidence.	Dashboard quality depends on source-data completeness.

These design choices reflect the thesis objective: the system must be technically executable, commercially understandable, and academically defensible. The design intentionally avoids presenting a single ML number as the answer. Instead, it presents a set of linked evidence objects: ERP price, should-cost, ML fair price, confidence, explanation, anomaly flag, cluster, and qualified savings.

5 Implementation Details

5.1 Repository Structure

File / folder	Purpose
app.py	Streamlit dashboard orchestration, page routing, and result display.
ctp/cost_model.py	Cost Digital Twin formula, should-cost components, and manual template adjustments.
ctp/erp_pipeline.py	ERP row validation, cleaning, supplier anonymization, and processed outputs.
ctp/market_data.py	Steel index and USD/INR FX source/fallback logic.
ctp/ml_models.py	Fair-label preparation, supervised models, confidence, SHAP/fallback explanation, anomaly detection, and clustering.
scripts/generate_ml_results.py	Standalone generation of model metrics, priced-part results, feature importance, cluster summary, and graphs.
tests/	Focused unit tests for cost model, ERP pipeline, and ML behaviour.

5.2 Prototype Capabilities

- Sheet-metal should-cost analysis using engineering attributes.
- Clickable part-level digital twin drill-down.
- ERP raw data upload and cleaning.
- Supplier anonymization and currency normalization.
- Live or fallback steel index and USD/INR FX adjustment.
- Portfolio-level ERP price versus predicted fair price comparison.
- Part-level cost breakdown and monthly ERP versus daily fair-price timeline.
- Manual should-cost template adjustments for scrap, rejection, tool maintenance, packing/forwarding, and tooling amortization.
- Linear Regression, Random Forest, XGBoost/fallback, SHAP/fallback explanation, Isolation Forest, and K-Means.
- Supplier benchmarking and geographic landed-cost comparison.
- Deployment readiness through Streamlit Community Cloud.

5.3 Run and Test Commands

Run command

```
streamlit run app.py
```

Result generation

```
python3 scripts/generate_ml_results.py
```

Validation

```
pytest -q
```

6 Intermediate Results and Evaluation Plan

Table 8: Current prototype summary metrics

Result	Value
Parts evaluated	30.0000
Prediction-ready parts	30.0000
Isolation Forest anomalies	13.0000
K-Means clusters	3.0000
SHAP explanation coverage percent	100.0000
Best training R2	1.0000
Lowest training MAE	0.7071
Lowest training RMSE	0.9055

Table 9: Model training metrics

Algorithm	MAE	MAPE	RMSE	R2	Backend
Linear Regression	109.48	15.0065	152.73	0.9906	Linear Regression
Random Forest	140.17	7.9557	318.24	0.9591	Random Forest
XGBoost	0.71	0.1124	0.91	1.0000	GradientBoosting fallback; install libomp for native XGBoost on macOS

Interpretation

The current metrics are prototype training metrics on a small demonstration set. Final thesis evaluation requires a larger real ERP-plus-drawing dataset and time-based validation.

Table 10: Top savings-opportunity examples

Part ID	Part name	ERP price	ML fair price	Savings	Confidence
SM-1005	Sensor guard bracket	INR 515.00	INR 449.95	INR 624440.41	Low
SM-1006	Base mounting plate	INR 575.00	INR 502.28	INR 588998.97	Low
SM-1024	Conveyor transfer chute	INR 7450.00	INR 7049.45	INR 380523.17	Low
SM-1028	Motor base welded assembly	INR 2410.00	INR 2295.52	INR 240405.97	Low
SM-1027	Terminal box cover	INR 365.00	INR 348.56	INR 151233.39	Low
SM-1026	Robot pedestal plate	INR 4825.00	INR 4686.18	INR 124940.64	Low
SM-1007	Electrical enclosure panel	INR 945.00	INR 933.90	INR 56605.62	Medium
SM-1011	Operator console side panel	INR 1035.00	INR 1023.55	INR 50372.74	Medium
SM-1018	Hydraulic manifold plate	INR 1250.00	INR 1237.35	INR 44280.33	Low
SM-1010	Drive cabinet back plate	INR 820.00	INR 812.84	INR 40122.76	Low

6.1 Result Interpretation

The intermediate results demonstrate that the implemented pipeline is executable end-to-end. The current data contains 30 demonstration engineering parts and 90 cleaned ERP rows from 97 imported source records. The system produces should-cost, ML predicted fair price, qualified savings, anomaly flags, cluster labels, confidence indicators, and explanation fields for the portfolio.

The qualified savings value must be interpreted carefully. The system counts savings only when the ML predicted fair price is below the ERP/current supplier price. This avoids overstating savings when the model predicts that the current supplier price is already lower than the fair-price estimate. As a result, the savings field is a conservative prioritization signal rather than a direct claim of immediate realizable savings.

The anomaly results should also be interpreted as review triggers rather than automatic error labels. Isolation Forest identifies unusual patterns in the feature space, but a procurement expert must still review whether the unusual pattern is caused by a real commercial issue, drawing complexity, one-time tooling, freight, emergency purchase conditions, or a valid supplier-specific reason.

The clustering output supports benchmark analysis by grouping similar parts. It helps the buyer compare parts with comparable cost structures instead of comparing all parts in one portfolio-level list. This is important because sheet-metal components can vary significantly by material, size, weight, bend count, hole count, finish, volume, and supplier region.

6.2 Discussion of Current Evidence

The current prototype provides implementation evidence but not final thesis validation. It proves that the architecture, data flow, costing logic, ML models, explainability layer, and dashboard outputs are operational. The remaining research step is to increase the amount of real ERP-linked engineering data and use that dataset for time-based validation and expert/reference template comparison.

The strongest evidence in the current stage is functional completeness. ERP cleaning, supplier anonymization, market adjustment, Cost Digital Twin calculation, ML fair-price prediction, explanation, anomaly detection, segmentation, supplier benchmarking, and result export all exist in the codebase. The weakest evidence is data representativeness, because real purchase history is available but drawing-level attributes are still prototype-level for the current report.

This distinction is important for thesis defense. The project should not claim that the current model is fully validated for industrial deployment. Instead, it should claim that the end-to-end methodology and prototype are implemented, and that the next stage will test robustness using expanded real data and expert validation.

6.3 Real Data Expansion Plan

The next thesis stage will focus on expanding the real dataset. More ERP rows alone are not sufficient; the ERP records must be connected to part-level engineering attributes for the same part IDs. Without that join, the system can clean purchase history but cannot fully evaluate whether price differences are caused by material, geometry, weight, finish, process complexity, or supplier-region effects.

Data area	Target	Purpose
ERP transactions	100+ rows minimum; 200-500 preferred	Part ID, supplier, country, PO date, quantity, unit price, currency, tooling/freight/emergency flags.
Part coverage	30+ real part IDs minimum; 50-100 preferred	Enough part diversity to reduce overfitting to a small demonstration set.
Drawing attributes	Same part IDs as ERP rows	Material, grade, thickness, length, width, weight, bends, holes, finish, and process.
Supplier coverage	5+ suppliers preferred	Needed for supplier benchmark and regional comparison.
Time span	6-12 months minimum; 12-24 months preferred	Needed for time-based train/test validation and market movement analysis.
Expert validation	10-20 manually checked parts	Needed to validate generated should-cost templates against reference values.

The real-data collection workbook prepared for the project captures ERP transactions, drawing attributes, supplier and market context, expert should-cost checks, and result-run tracking. Once populated, the same implemented pipeline can be rerun to compare updated results with the current prototype baseline.

6.4 Validation Strategy

Validation layer	Measure	Reason
Data completeness	ERP row count, engineering join rate, missing critical fields	Ensures the dataset is strong enough for modelling.
Cost-template correctness	Field-level match, assumption match, total should-cost variance	Checks generated should-cost against expert/reference templates.
Prediction accuracy	MAE, MAPE, RMSE, R2	Measures fair-price prediction quality.
Commercial usefulness	Accepted savings opportunities, rejected false positives, buyer comments	Checks whether outputs support procurement decisions.
Explainability usefulness	Reviewer rating of top-driver explanation	Checks whether the model explanation is usable in negotiation.

The evaluation strategy is intentionally layered. A regression metric alone cannot prove procurement usefulness. The system must also show that its generated should-cost fields are correct, that its anomaly flags are meaningful, that its savings logic is conservative, and that explanations are understandable to procurement users.

6.5 Evaluation Plan

Data Collection	Cleaning	Model Rerun	Metric Evaluation	Expert Validation	Thesis Result
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Figure 6: Validation workflow for next thesis stage

Table 11: Evaluation metrics

Area	Metric	Purpose
Regression	MAE, MAPE, RMSE, R2	Compare predicted fair price with cleaned fair-price labels or expert reference.
Anomaly detection	Precision, recall, F1 where labels exist	Evaluate flagged cases against procurement/expert review.
Procurement impact	Price gap, qualified savings, supplier benchmark	Measure commercial usefulness.
Template correctness	Field accuracy, completeness, cost variance	Compare generated template against expert/reference template.
Explainability	Top-driver usefulness	Check whether explanation supports negotiation and review.

6.6 Current Limitation

The current implementation includes real ERP purchase history imported into the app schema and cleaned into usable processed records. However, drawing-level engineering attributes in the prototype engineering master are not yet fully replaced by real CAD/OCR-derived fields. This limitation affects final model generalization and will be addressed by collecting more real data and matching drawing attributes before final thesis evaluation.

7 Conclusion and Future Work

This thesis phase demonstrates a working explainable procurement intelligence prototype for sheet-metal sourcing. The system integrates ERP cleaning, Cost Digital Twin modelling, ML fair-price prediction, SHAP/fallback explanation, Isolation Forest anomaly detection, K-Means segmentation, supplier benchmarking, geographic cost comparison, and Streamlit-based decision support.

The main research contribution is the integration of procurement history, engineering cost drivers, market context, and explainable AI into a traceable fair-price decision pipeline. This creates a stronger basis for procurement negotiation than using ERP purchase history alone.

7.1 Future Work

- Collect a larger real ERP purchase-history dataset for sheet-metal parts.
- Replace prototype engineering attributes with real drawing/CAD/OCR-derived attributes.
- Join transaction-level ERP history with part-level engineering master data.
- Run time-based train/test validation using older and newer transaction records.
- Validate selected should-cost templates against expert/reference cost sheets.
- Add cached market-data history for time-aware fair-price trend modelling.
- Improve deployment access and demonstration readiness.

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Appendix A: Dataset Schema

data/sample_parts.csv

The file contains 30 rows and 25 columns.

Table 12: Dataset schema for data/sample_parts.csv

No.	Column	Purpose
1	part_id	Part identity or part-level descriptor.
2	part_name	Part identity or part-level descriptor.
3	category	Supporting input or derived output field.
4	material	Engineering cost-driver field.
5	material_grade	Engineering cost-driver field.
6	thickness_mm	Engineering cost-driver field.
7	length_mm	Supporting input or derived output field.
8	width_mm	Supporting input or derived output field.
9	weight_kg	Engineering cost-driver field.
10	bend_count	Engineering cost-driver field.
11	hole_count	Engineering cost-driver field.
12	surface_finish	Engineering cost-driver field.
13	finish_cost_per_part	Part identity or part-level descriptor.
14	material_rate_per_kg	Engineering cost-driver field.
15	cycle_time_min	Supporting input or derived output field.
16	energy_kwh_per_part	Part identity or part-level descriptor.
17	energy_rate_per_kwh	Supporting input or derived output field.
18	labour_hours	Supporting input or derived output field.
19	labour_rate_per_hour	Supporting input or derived output field.
20	overhead_pct	Supporting input or derived output field.
21	supplier_margin_pct	Supplier, sourcing, or regional-cost context.
22	current_supplier	Supplier, sourcing, or regional-cost context.
23	supplier_region	Supplier, sourcing, or regional-cost context.
24	erp_price	Pricing, cost, variance, or savings calculation field.
25	annual_volume	Supporting input or derived output field.

data/erp_raw_sample.csv

The file contains 97 rows and 14 columns.

Table 13: Dataset schema for data/erp_raw_sample.csv

No.	Column	Purpose
1	Part Number	Part identity or part-level descriptor.
2	Part Description	Part identity or part-level descriptor.
3	Category	Supporting input or derived output field.
4	Supplier Name	Supplier, sourcing, or regional-cost context.
5	Supplier Country	Supplier, sourcing, or regional-cost context.
6	PO Date	Time field for trend and validation analysis.
7	Quantity	Supporting input or derived output field.
8	Unit Price	Pricing, cost, variance, or savings calculation field.
9	Currency	Supporting input or derived output field.
10	Total Purchasing Price Local	Pricing, cost, variance, or savings calculation field.
11	Original Country Code	Supplier, sourcing, or regional-cost context.
12	Bracket Type	Supporting input or derived output field.
13	Material Process Tag	Engineering cost-driver field.
14	Tooling Flag	Supporting input or derived output field.

data/processed/erp_cleaned.csv

The file contains 90 rows and 13 columns.

Table 14: Dataset schema for data/processed/erp_cleaned.csv

No.	Column	Purpose
1	part_id	Part identity or part-level descriptor.
2	description	Supporting input or derived output field.
3	category	Supporting input or derived output field.
4	supplier_id	Supplier, sourcing, or regional-cost context.
5	country	Supplier, sourcing, or regional-cost context.
6	region	Supplier, sourcing, or regional-cost context.
7	po_date	Time field for trend and validation analysis.
8	quantity	Supporting input or derived output field.
9	unit_price	Pricing, cost, variance, or savings calculation field.
10	currency	Supporting input or derived output field.
11	unit_price_usd	Pricing, cost, variance, or savings calculation field.
12	year	Time field for trend and validation analysis.
13	month	Time field for trend and validation analysis.

outputs/ml_results/ml_priced_parts_results.csv

The file contains 30 rows and 99 columns.

Table 15: Dataset schema for outputs/ml_results/ml_priced_parts_results.csv

No.	Column	Purpose
1	part_id	Part identity or part-level descriptor.
2	part_name	Part identity or part-level descriptor.
3	category	Supporting input or derived output field.
4	material	Engineering cost-driver field.
5	material_grade	Engineering cost-driver field.
6	thickness_mm	Engineering cost-driver field.
7	length_mm	Supporting input or derived output field.
8	width_mm	Supporting input or derived output field.
9	weight_kg	Engineering cost-driver field.
10	bend_count	Engineering cost-driver field.
11	hole_count	Engineering cost-driver field.
12	surface_finish	Engineering cost-driver field.
13	finish_cost_per_part	Part identity or part-level descriptor.
14	material_rate_per_kg	Engineering cost-driver field.
15	cycle_time_min	Supporting input or derived output field.
16	energy_kwh_per_part	Part identity or part-level descriptor.
17	energy_rate_per_kwh	Supporting input or derived output field.
18	labour_hours	Supporting input or derived output field.
19	labour_rate_per_hour	Supporting input or derived output field.
20	overhead_pct	Supporting input or derived output field.
21	supplier_margin_pct	Supplier, sourcing, or regional-cost context.
22	current_supplier	Supplier, sourcing, or regional-cost context.
23	supplier_region	Supplier, sourcing, or regional-cost context.
24	erp_price	Pricing, cost, variance, or savings calculation field.
25	annual_volume	Supporting input or derived output field.
26	blank_weight_kg	Engineering cost-driver field.
27	scrap_weight_kg	Engineering cost-driver field.
28	scrap_rate_per_kg	Supporting input or derived output field.
29	scrap_recovery_pct	Supporting input or derived output field.
30	rejection_pct	Supporting input or derived output field.
31	tool_maintenance_pct	Supporting input or derived output field.
32	packing_forwarding_pct	Supporting input or derived output field.
33	tooling_cost	Pricing, cost, variance, or savings calculation field.
34	tooling_cost_per_part	Part identity or part-level descriptor.
35	blank_area_m2	Supporting input or derived output field.
36	grade_factor	Supporting input or derived output field.
37	thickness_factor	Engineering cost-driver field.

No.	Column	Purpose
38	market_material_rate_per_kg	Engineering cost-driver field.
39	raw_material_cost_gross	Pricing, cost, variance, or savings calculation field.
40	scrap_recovery	Supporting input or derived output field.
41	material_cost	Pricing, cost, variance, or savings calculation field.
42	predicted_energy_rate_per_kwh	Supporting input or derived output field.
43	energy_cost	Pricing, cost, variance, or savings calculation field.
44	predicted_labour_rate_per_hour	Supporting input or derived output field.
45	labour_cost	Pricing, cost, variance, or savings calculation field.
46	machine_rate_per_hour	Supporting input or derived output field.
47	operation_minutes	Supporting input or derived output field.
48	bend_cost	Pricing, cost, variance, or savings calculation field.
49	piercing_cost	Pricing, cost, variance, or savings calculation field.
50	process_complexity_cost	Pricing, cost, variance, or savings calculation field.
51	surface_finish_rate_per_m2	Engineering cost-driver field.
52	surface_finish_cost	Pricing, cost, variance, or savings calculation field.
53	conversion_cost	Pricing, cost, variance, or savings calculation field.
54	predicted_overhead_pct	Supporting input or derived output field.
55	overhead	Supporting input or derived output field.
56	rejection_allowance	Supporting input or derived output field.
57	tool_maintenance_cost	Pricing, cost, variance, or savings calculation field.
58	packing_forwarding_cost	Pricing, cost, variance, or savings calculation field.
59	tooling_amortization_cost	Pricing, cost, variance, or savings calculation field.
60	manual_template_adjustment_cost	Pricing, cost, variance, or savings calculation field.
61	cost_before_margin	Pricing, cost, variance, or savings calculation field.
62	predicted_supplier_margin_pct	Supplier, sourcing, or regional-cost context.
63	supplier_margin	Supplier, sourcing, or regional-cost context.
64	should_cost	Pricing, cost, variance, or savings calculation field.
65	price_gap	Pricing, cost, variance, or savings calculation field.
66	price_gap_pct	Pricing, cost, variance, or savings calculation field.
67	savings_opportunity	Pricing, cost, variance, or savings calculation field.
68	annual_opportunity	Supporting input or derived output field.
69	opportunity_status	Supporting input or derived output field.
70	gap_status	Supporting input or derived output field.
71	commodity_index	Supporting input or derived output field.
72	fx_rate	Supporting input or derived output field.
73	critical_missing_fields	Supporting input or derived output field.
74	optional_missing_fields	Supporting input or derived output field.
75	prediction_readiness	Supporting input or derived output field.
76	raw_erp_price_label	Pricing, cost, variance, or savings calculation field.
77	erp_to_should_cost_gap_pct	Pricing, cost, variance, or savings calculation field.

No.	Column	Purpose
78	ml_fair_price_label	Pricing, cost, variance, or savings calculation field.
79	label_quality_status	ML label, explanation, confidence, or feature-analysis field.
80	training_sample_weight	Engineering cost-driver field.
81	similar_erp_history_count	Supporting input or derived output field.
82	prediction_confidence	ML label, explanation, confidence, or feature-analysis field.
83	market_source_status	Supporting input or derived output field.
84	linear_regression_fair_price	Pricing, cost, variance, or savings calculation field.
85	random_forest_fair_price	Pricing, cost, variance, or savings calculation field.
86	xgboost_fair_price	Pricing, cost, variance, or savings calculation field.
87	ai_predicted_fair_price	Pricing, cost, variance, or savings calculation field.
88	ai_price_gap	Pricing, cost, variance, or savings calculation field.
89	ai_price_gap_pct	Pricing, cost, variance, or savings calculation field.
90	ai_savings_opportunity	Pricing, cost, variance, or savings calculation field.
91	should_cost_variance	Pricing, cost, variance, or savings calculation field.
92	should_cost_variance_pct	Pricing, cost, variance, or savings calculation field.
93	isolation_forest_score	Unsupervised analytics output field.
94	isolation_forest_flag	Unsupervised analytics output field.
95	kmeans_cluster	Unsupervised analytics output field.
96	shap_top_feature	ML label, explanation, confidence, or feature-analysis field.
97	shap_top_feature_impact	ML label, explanation, confidence, or feature-analysis field.
98	shap_explanation_method	ML label, explanation, confidence, or feature-analysis field.
99	shap_procurement_explanation	ML label, explanation, confidence, or feature-analysis field.

Appendix B: Algorithm and Validation Support

ERP Cleaning Pipeline

1. Load raw ERP file.
2. Validate mandatory columns.
3. Remove duplicates and invalid rows.
4. Normalize dates, quantity, price, and currency fields.
5. Map country to supplier region.
6. Anonymize supplier names.
7. Write cleaned ERP output and quality report.

Cost Digital Twin

1. Read part engineering attributes.
2. Calculate gross material cost and scrap recovery.
3. Apply market-adjusted steel rate.
4. Estimate energy, labour, bend, hole, and finish cost.
5. Apply overhead and manual template adjustments.
6. Apply supplier margin.
7. Return should-cost and cost breakdown.

ML Fair-Price Prediction

1. Prepare cleaned fair-price labels.
2. Train baseline and non-linear models.
3. Select primary model output.
4. Calculate ML price gap and savings opportunity.
5. Generate confidence and explanation fields.

Anomaly Detection

1. Create feature matrix from cost, ERP, market, and part attributes.
2. Standardize/prepare numeric fields.
3. Run Isolation Forest.
4. Assign anomaly score and flag.
5. Prioritize flagged records for procurement review.

Part Segmentation

1. Create comparable-part feature matrix.
2. Run K-Means clustering.
3. Assign part cluster.
4. Summarize cluster-level spend and price gaps.
5. Use clusters for benchmark review.

Validation Support

Validation area	Metric	Evidence
Regression accuracy	MAE, MAPE, RMSE, R2	Time-based split after real ERP + drawing data collection.
Should-cost correctness	Field-level match and total variance	Compare generated template with expert/reference template.
Anomaly usefulness	Expert review of top flagged records	Treat anomaly flag as review trigger, not proof of error.
Savings reliability	Qualified savings business rule	Savings counted only when ML fair price is lower than ERP price.
Explainability usefulness	Top-driver clarity	Check whether explanation helps procurement negotiation.